Effect of Path Length on Robustness in Power Grids

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Abstract

If a lot of renewable energy is introduced in power grids, non-small fluctuations of power consumption in the loads can emerge. In such a situation, it is important to clarify how robust each node is against non-small fluctuations, which has not been fully considered in the classical electrical power engineering. We consider non-small fluctuations of effective power in one load node and propose a new concept of node-wise robustness of power grids. We define the robustness as the threshold value of the effective power, above which the steady state loses its stability. In a network model with the topology of the power grid in eastern Japan, we find numerically that the shortest path length from generators is most highly correlated with the robustness of the load nodes. The mechanism for the correlation is discussed using a simple network.

Keyword: Power Grid, Robustness, Path Length

1 Introduction

The output of renewable energy connected to end-users can be regarded as the negative power consumption in loads. Therefore, if a significant amount of renewable energy is introduced in power grids, the fluctuations of renewable energy cause the variation of power consumption in the loads. We consider non-small fluctuations of effective power in one load node and propose a new concept of node-wise robustness of power grids [1].

2 Model

We adopt the mathematical model proposed in Ref. [2]. In this model, the behavior of generators is described using the swing equations. On the other hand, the behavior of loads is determined with the power flow calculations.

3 Results

We derive the topology of the power grid from the annual reports published by Tokyo Electric Power Company Inc.[1] and Tohoku Electric Power Company Inc.[2]

We find numerically that the smaller shortest path length from generators is, the more the node-wise robustness is [1]. This is probably because generators can adjust the influence of fluctuations in power consumption by feeding the effective power. The mechanism for the correlation is discussed using a simple network.

Acknowledgements

This research was supported by JST, CREST.

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